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Swing Phase Control of Semi-Active Prosthetic Knee using Neural Network Predictive Control with Particle Swarm Optimization

Kittipong Ekkachai and Itthisek Nilkhamhang

***Abstract*—In recent years, intelligent prosthetic knees have been developed that enable amputees to walk as normally as possible when compared to healthy subjects. Although semi-active prosthetic knees utilizing magnetorheological (MR) dampers offer several advantages, they lack the ability to generate active force that is required during some states of a normal gait cycle. This prevents semi-active knees from achieving the same level of performance as active devices. In this work, a new control algorithm for a semi-active prosthetic knee during the swing phase is proposed to reduce this gap. The controller uses neural network predictive control and particle swarm optimization to calculate suitable command signals. Simulation results using a double pendulum model show that the generated knee trajectory of the proposed controller is more similar to the normal gait than previous open-loop controllers at various ambulation speed. Moreover, the investigation shows that the algorithm can be calculated in real-time by an embedded system, allowing for easy implementation on real prosthetic knees.**

***Index Terms*—prosthetic knee, magnetorheological (MR) damper, neural network predictive control, particle swarm optimization.**

I. Introduction

PEOPLE with leg amputation caused by accidents, diseases, or congenital disorders usually have reduced movement ability. A prosthetic leg is needed to replace the lost limb and improve their quality of life. The aim is to help amputees carry on their daily activities such as walking or standing like healthy people. An above-knee (transfemoral) amputation is a type of leg amputation that occurs between the knee and the hip. The prosthetic device for above-knee amputee must include a knee joint that is the most important part of the leg. This prosthetic knee can be divided into three types: mechanically passive, variable-damping or semi-active, and powered or active prosthetic knee. The mechanically passive knee does not require a power supply to operate. It generates knee damping that is suited for only one walking speed. Variable-damping knee gives advantages over mechanically passive knee, such as enhanced knee stability, adaptation to changing environment, and ability to change ambulatory speed. It improves gait symmetry, which decreases energy consumption [1]–[3] and may lead to a reduction in degenerative musculoskeletal changes for the amputee [4]. Examples of commercial variable-damping prosthetic knee include Otto Bock C-leg and Össur Rheo Knee. For the last type, powered or active prosthetic knee has a mechanism to generate positive power and damping force. It makes the amputee capable of activities such as walking up staircases and standing from a sitting position. However, this kind of prosthetic knee consumes high level of energy and requires a large actuator that increases the weight and reduces battery life. Many works on active knee development have been proposed [5]–[7]. However, only one product (Össur Power Knee) has been commercialized.

The walking cycle consists of stance and swing phase that are divided by heel-strike and toe-off events. From the previous study, different knee torque is required for various walking speed during the swing phase [8]. During the swing phase of the normal gait, the knee requires both active and damping forces that are generated by muscles. As shown in Fig. 1, knee torque and knee angular velocity during swing phase of the normal gait are plotted based on the data from [9]. If the knee angular velocity has the same sign as the knee torque, then the knee muscles act like a damper. Otherwise, the knee muscles act like a powered actuator. From start to 25% of the swing phase, the knee generates damping force that provides friction to limit the maximum knee flexion. From 25% to 40% of the swing phase, the knee generates force required for extension. From 40% to 95% of the swing phase, the knee again generates friction. In the terminal swing, active force is required to flex the knee to prepare for shock absorption upon impact at heel-strike. From these behaviors, an active prosthetic knee will be better at mimicking knee kinematics of the normal gait than semi-active or passive knee because it can generate both active and damping forces while the semi-active knee can generate only damping force. The swing phase simulation which uses the data from [9] is shown in Fig. 2. The result shows that the simulated knee providing only damping force cannot generate knee trajectory of the normal gait. This conclusion is similar to a previous research that compares passive and active knees in the swing phase [10]. In order to achieve performance comparable to active knee, an intelligent control algorithm for semi-active knee is required.

Magnetorheological (MR) fluid damper is a semi-active device that is widely used in variable-damping knees. It is a cylinder or rotary disc containing MR fluid that can change its characteristics between liquid and semi-solid by applying different strength of magnetic field through varying voltage or

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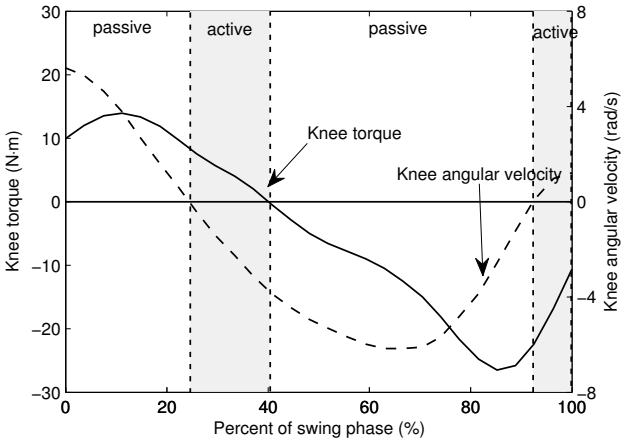


Fig. 1. Knee torque and angular velocity during swing phase of normal gait [9]. The figure shows that the knee requires both active and damping forces depending on the period within the swing phase.

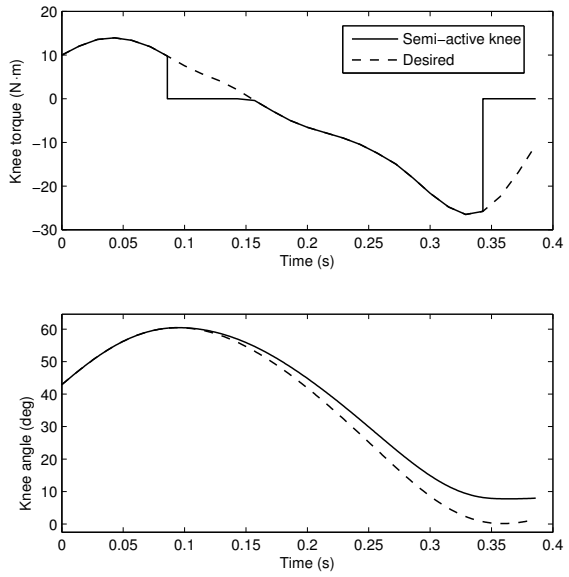


Fig. 2. (Top) The torque generated from semi-active knee is compared with the desired torque. (Bottom) The knee angle simulated by the generated torque profile is compared with the knee angle of the normal gait using data from [9].

current. It is an energy efficient device with fast response time. Some variable-damping knees use MR damper to generate damping force which is controlled by a microcontroller. A repetitive controller in conjunction with PD and a computed torque control laws is introduced in [11]. It calculates desired torque from a mechanical dynamic model and then uses a PD controller to control that torque. The controlled torque is adapted due to the error of the previous step by a repetitive control law. The experiment in [11] shows that the controller will give a good performance after passing 4 steps. User-adaptive control algorithm is introduced to control MR damper prosthetic knee [8]. The damping constant in each state will be changed if the knee trajectory does not meet the criteria. In swing phase, the maximum knee flexion is limited to 70 degree. If the maximum knee flexion exceeds 70 degree, the controller will increase the damping constant that will be used in the next step. Their clinical results show that the knee can improve gait symmetry in some cases but power generation ability is still required for the remaining participants. In [12], adaptive modular active leg (AMAL) is proposed which includes a method to calculate desired knee angle by using the central pattern generator and fuzzy-logic. In each walking cycle, the control period is divided into 8 equal intervals that use different control voltages. Predefined constant values are used as the control voltages in each interval. The AMAL controller uses hip data to determine the state of the walking cycle. These swing phase control algorithms for MR damper prosthetic knees are open-loop control strategies that determine a series of input voltages before each swing period and contain some mechanisms to adjust the signal during the next step if the generated knee angle does not match with the desired knee angle [8], [12]. The main disadvantage is that these methods assume a constant walking speed and are slow to adapt to changes in gait. In addition, ideal swing phase control requires both active and passive force, while the MR damper can only produce passive force. Thus, a suitable command voltage for generating a normal swing trajectory is difficult to determine. To overcome this obstacle, the swing phase control must contain look-ahead calculations for the damping force so that the effects of the current voltage signal will appear during the next control cycle.

This research proposes a novel algorithm using model predictive control (MPC) for swing phase control of an MR damper prosthetic knee, with the ability to predict future knee angles and operate at various ambulation speeds. MPC is an excellent algorithm for process control, and is widely used in power plants, petroleum refineries, chemicals and food processing, automotive, and aerospace applications [13]. MPC has the ability to predict future events and take control actions accordingly, while PID and LQR controllers do not have this predictive ability. For variable-damping prosthetic knee controller, MPC provides the ability to generate damping force in advance for compensating the force required at the active period of the swing phase. Therefore, MR damper prosthetic knee using MPC would generate prosthetic knee trajectory compared to normal gait and has the same benefits as active prosthetic knee, while retaining the advantages of variable-damping prosthetic knee. In this investigation, neural network predictive control (NNPC), a kind of MPC that represent the plant with neural network [14], is proposed as the swing phase control. The proposed controller utilizes a neural network model to estimate future responses of a prosthetic knee in order to find the best control signal. Particle swarm optimization (PSO) is employed to determine the optimal control signal during the swing phase [15]. This method is suited for problems involving large search space with multiple local optimums, such as in this research. Moreover, a suitable stopping criteria can be specified so that PSO finishes within the current control cycle. This control algorithm is designed to support various walking speeds and has been tested by simulation in comparison to user-adaptive control [8]. A double pendulum model

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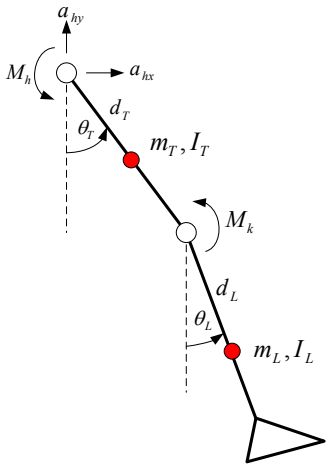


Fig. 3. The double pendulum model used to simulate swing phase of prosthetic knee.

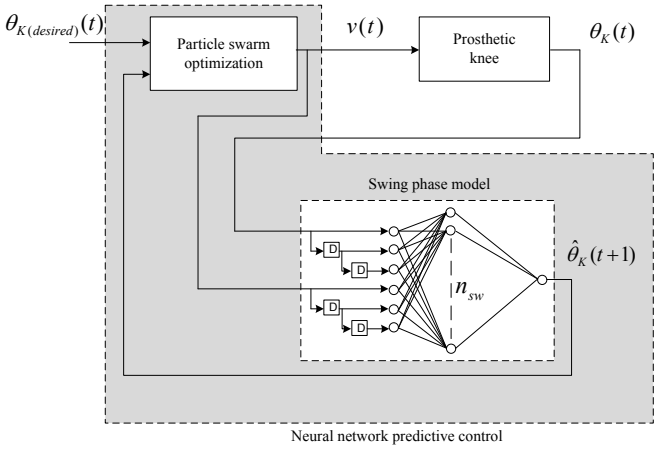


Fig. 4. The block diagram of NNPC.

is used to simulate the prosthetic leg via the SimMechanic Toolbox of MATLAB. All datasets were captured from the walking patterns of a male subject with normal gait. The parameters of the MR damper attached to the prosthetic knee were obtained by experimental identification. The results show that the proposed algorithm performs well at various walking speeds, is robust to changing system parameters, and can be implemented on real-time embedded platforms.

II. Double Pendulum Model

The swing phase of the transfemoral prosthetic knee is simulated by a double pendulum model which was proposed by [16], [17]. It includes two links representing the thigh and leg, consisting of shank and foot, as shown in Fig. 3. The torque generated at the hip and knee joints during swing can be calculated by the following equation:

$$\begin{aligned} M_K =& I_L\alpha_L + m_L d_L(L_T\alpha_T\cos(\theta_L-\theta_T) \\ &+ L_T\omega_T^2\sin(\theta_L-\theta_T) \\ &+ a_{hx}\cos\theta_L + (a_{hy}+g)\sin\theta_L), \end{aligned} \tag{1}$$

$$\begin{aligned} M_H =& M_K + (I_T + m_L L_T^2)\alpha_T \\ &+ m_L L_T d_L(\alpha_L\cos(\theta_L-\theta_T) - \omega_L^2\sin(\theta_L-\theta_T)) \\ &+ (a_{hx}\cos\theta_T + (a_{hy}+g)\sin\theta_T)(m_T d_T + m_L L_T), \end{aligned} \tag{2}$$

where M_K and M_H are the knee and hip torques, respectively. I, m, L and d are moment of inertia at center of mass, segment mass, segment length, proximal length of center of mass, respectively. θ, ω, α are the angle, angular velocity and angular acceleration, respectively. The subscripts T and L denote the thigh and leg. a_{hx} and a_{hy} are the linear acceleration of the hip along the x- and y-axes, where g is the gravitational constant at 9.81 m/s^2. The knee angle in this model (θ_K) can be calculated by the following equation:

$$\theta_K = \theta_T - \theta_L, \tag{3}$$

where θ_T is the thigh angle and θ_L is the leg angle.

A cylinder-type MR damper is utilized as a knee actuator. It is attached at distance d_{MR} away from the knee joint. The knee trajectory can be controlled with variable knee torque generated by the MR damper. The force generated by the MR damper can be converted to knee torque depending on the attachment position:

$$M_K = d_{MR}\cdot|F|\cos(\theta_K), \tag{4}$$

where F is the damping force generated by the MR damper and d_{MR} is the distance between center of knee joint and MR damper attachment position.

III. Neural Network Predictive Control

Details of NNPC algorithm, which is proposed to control prosthetic knee during swing period, are described in this section. The algorithm is applied to calculate a suitable command voltage for a MR damper attached to a prosthetic knee to generate torque. The block diagram of NNPC is shown in Fig. 4, consisting of the swing phase model and particle swarm optimization (PSO). NNPC requires a desired knee angle and a measurement of actual knee angle as the inputs of controller. The method to acquire these data is beyond to the scope of this work; however, it have been found in [7], [12], [18]. As shown in Fig. 4, the input of the system consist of two data which are desired knee angle ($\theta_{K(\text{desired})}$) and actual knee angle (θ_K) that is required as a feedback signal. The swing phase model is implemented with feedforward neural network (FNN) which receives two input data: the actual knee angle and the voltage for predicting future knee angle. At each control interval, PSO is used to find an optimal voltage over the prediction horizon. The first command value of the optimal voltages is employed to control the prosthetic knee. During each control interval, the actual knee angle data is fed to NNPC to achieve close-loop control.

A. Swing phase model

In order to implement a predictive control strategy, NNPC requires a method for estimating the future knee angle trajectory when a certain command voltage is applied. Thus, a

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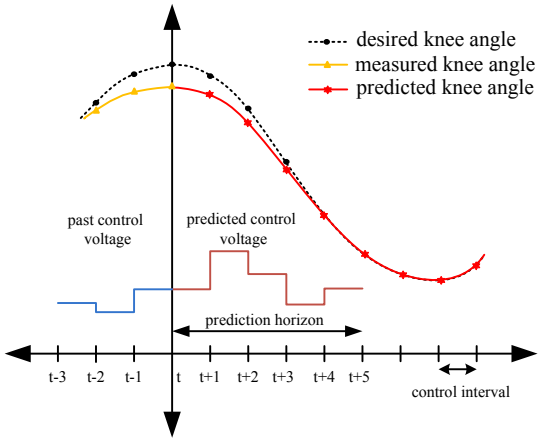


Fig. 5. The work of Neural Network Predictive Control (NNPC).

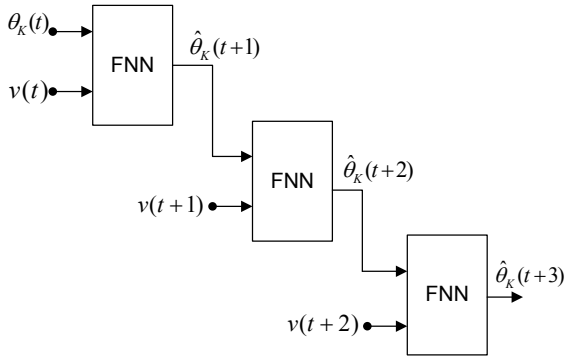


Fig. 6. Neural network calculation in prediction periods.

swing phase model is included in NNPC to accomplish this task. It is used to predict the knee angle during each iteration of the optimization process in order to determine the optimal command voltage. The control performance of NNPC depends on the accuracy of this swing phase model. In this work, the swing phase model is implemented with FNN, as shown in Fig. 4. It predicts the future knee angle ($\hat{\theta}_K(t+1)$) with two input variables which are knee angle (θ_K) and command voltage (v). The history of the inputs is also required by the FNN to capture dynamics of the system. The total number of FNN input data is 6. The network has one hidden layer with a sufficient number of nodes (n_{sw}) and employs a tangent sigmoid function. The output layer has one node that applies a linear transfer function to estimate future knee angle.

B. Predictive control

To determine a suitable command voltage, NNPC has a mechanism to simulate the behavior of the knee at any voltage levels. It utilizes the swing phase model for predicting the knee angle and optimization to calculate the command voltage. An example of the predictive control algorithm is shown in Fig. 5. At time t, an arbitrary control voltage and the measured knee angle are inputs of the swing phase model. The model predicts a knee angle at time $t+1$. Then, the algorithm continues to predict the knee angle at time $t+2$ by randomizing an arbitrary control voltage and applies the predicted knee angle from the previous interval as a new input. The output in the previous cycle is the input of the current cycle. The calculation during the prediction periods is shown in Fig. 6. The loop continues in this manner until it reaches the maximum prediction horizon. The predictive control optimizes these voltage signals and then selects one that gives the lowest error. Once the optimal voltage signal is found, only the first value of the optimal voltage signal is applied to control the prosthetic knee. The optimization function (J) in the predictive control is based on the square error according to the following equation:

$$J = \sum_{i=1}^{n_{ph}} w_i[\theta_{K(\text{desired})}(t+i) - \hat{\theta}_K(t+i)]^2, \tag{5}$$

where n_{ph} is the prediction horizon and w_i is the weighting coefficient which reflects the relative importance of the horizon numbers. In the proposed controller, the optimization function gives higher priority for latter horizon events. This is because the MR damper requires response time to generate damping force due to its characteristic. It should be noted that the prediction horizon affects the quality of control and the calculation time. When it is too small, the control quality is poor. In contrast, when the prediction horizon is too high, the calculation time is large due to the increased search space that must be optimized. In order to find a suitable prediction horizon, various numbers are tested in the simulation.

C. Particle swarm optimization

The optimization process utilizes PSO to optimize the voltage signal over the prediction horizon. The idea of PSO was inspired by bird flocking or fish schooling to share their collective knowledge for finding food [15]. Each swarm member or particle movement depends on the own experience and experiences of other particles, allow the swarm to move towards the global optimal position. The behavior of a particle in an iteration of PSO can be described by the following equation:

$$\begin{aligned} \boldsymbol{vel}(t+1) =& w_v \cdot \boldsymbol{vel}(t) + c_1 \cdot r_1 \cdot (\boldsymbol{p}(t) - \boldsymbol{x}(t)) \\ &+ c_2 \cdot r_2 \cdot (\boldsymbol{g}(t) - \boldsymbol{x}(t)), \end{aligned} \tag{6}$$

$$\boldsymbol{x}(t+1) = \boldsymbol{x}(t) + \boldsymbol{vel}(t+1), \tag{7}$$

where $\boldsymbol{x}$, $\boldsymbol{vel}$, $\boldsymbol{p}$ and $\boldsymbol{g}$ are position, velocity of a particle, the best position visited so far by the particle and the best position visited by all the particles, respectively. The variables w_v, c_1 and c_2 are weighting coefficients. r_1 and r_2 are the uniform random variables in the range (0,1). The bold variables represented in $1 \times n_{dm}$ vector and n_{dm} is the size of the space of a given problem, which in this case corresponds to the size of the prediction horizon ($n_{dm} = n_{ph}$).

In NNPC, each particle ($\boldsymbol{x}$) represents the sequence of voltages applied to the knee in the future over the prediction horizon, as described by the following equation:

$$\boldsymbol{x} = [v_t, v_{t+1}, v_{t+2}, ..., v_{t+n_{ph}-1}], \tag{8}$$

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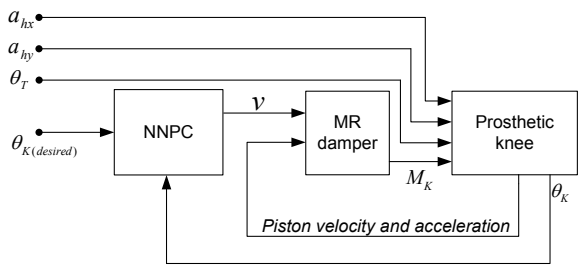


Fig. 7. The simulation system for the proposed swing phase controller performance verification.

TABLE I
SUMMARY INFORMATION OF PROSTHETIC LEG AND ACTUAL THIGH.

Parameter	Value	Unit
Body weight	83	kg
Body height	1.75	m
I_L	0.1592	kg·m^2
I_T	0.1291	kg·m^2
m_T	8.3	kg
m_L	3.0	kg
L_T	0.429	m
L_L	0.4985	m
d_T	0.1858	m
d_L	0.302	m

where v_t is an applied voltage at the current control period, and $v_{t+1}, v_{t+2}, ..., v_{t+n_{ph}-1}$ are the applied voltage in the future. This sequence of voltages will be fed to the swing phase model to predict a sequence of future knee angle, and then two inputs, $\theta_{K(\text{desired})}$ and $\hat{\theta}_K$, will be used to calculate the optimization function in Equation (5). Once PSO reaches to the stopping criteria, only the first command (v_t) in the best voltage sequence will be selected as the output of NNPC.

IV. SIMULATION

The performance of the proposed controller is evaluated by simulation for various walking speeds. The simulation system is constructed in MATLAB Simulink environment and consists of NNPC controller, MR damper model and simulated prosthetic knee, as shown in Fig. 7. The inputs of the system consist of desired knee angle, thigh angle and hip acceleration in x and y axis. These data are captured from the normal gait of a healthy participant. NNPC applies the desired knee angle, the thigh angle in conjunction with the measured knee angle to generate a suitable voltage command signal for the MR damper. The direct model of the MR damper is used to calculate the damping force. Then, this force is converted to the knee torque. The prosthetic knee model returns the measured knee angle and excitation data to NNPC and the MR damper model, respectively. The details of each component development are described in the following subsections.

A. Nomal gait data acquisition

Normal gait data is collected from a male subject who weighs 83 kg and 1.75 m tall. The data along the sagittal plan is recorded while the man is walking on a treadmill at speed of 2.4, 3.0, 3.6, 4.2, 4.8 and 5.4 km/h. He wears a bodysuit with three markers attached at hip, knee and ankle joints. A high-speed camera (Point Grey FL3-U3-13Y3M) with a resolution of 1280 x 1024 pixels and capable of recording 150 frames per second is fitted with a Fujinon 4-15.2 mm f/1.5 lens and used to record the lower body. Image Processing Toolbox in MATLAB is applied to implement a program that extracts the three joint coordinates, which are converted to thigh angle (θ_T), leg angle (θ_L), knee angle (θ_K), hip acceleration in x-axis (a_{hx}) and hip acceleration in y-axis (a_{hy}), as shown in Fig. 3. The system records 5 steps after the subject achieves steady-state gait at each walking speed, and determines the average data. These data will be used for training the swing phase model and validating the performance of NNPC. The example of knee angles and raw images that were captured from the system is shown in Fig. 8. This gait cycle can be divided into 2 states, which are stance and swing phase. Stance phase starts at heel-strike until toe-off and swing phase starts from toe-off until heel-strike. The knee angles at swing phase at different walking speeds are shown in Fig. 9. The results show that the maximum knee flexion is increased when the walking speed increases.

B. Prosthetic knee simulation

The prosthetic knee during swing phase is modeled using the double pendulum model constructed by SimMechanics Toolbox in MATLAB Simulink. The model includes two rigid bodies (thigh and leg) and two joints. The thigh data is determined from the actual thigh of the participant using the ratio investigated by [9]. The weight of the thigh is approximately one tenth of the body weight and the center of mass is located away from the hip by approximately 0.433 of the thigh length. The leg information determined by a prosthetic limb consists of knee, shank and foot. It is assumed that the total weight of the limb is 3 kg. The information of the prosthetic leg and the actual thigh are used in the simulation and shown in Table I. The simulation model has two joints, which are the hip and knee joints. The hip joint is controlled by the amputee's body, while the knee joint is controlled by a prosthetic knee controller. The hip joint rotation and movement mimics actual hip trajectories of the participant at various walking speeds. The MR damper attaches to the knee joint and is located at the shank of the prosthetic leg with a distance of 3 cm from the knee joint. A model of MR damper is applied to calculate the damping force generated by the MR damper. The knee torque (M_K) can be calculated from (4) with $d_{MR} = 0.03$ m (the distance between knee joint and MR damper attachment position). The prosthetic knee plant has a constraint that the knee angle cannot go below 0 degree. The sample time of the simulation is 1 ms. The generated knee angle, piston velocity and piston acceleration from the prosthetic knee are used as feedback signals for NNPC and MR damper model.

C. Magnetorheological damper model

A modified EHM-based FNN model is selected to represent MR damper in the simulation. The original EHM-based FNN

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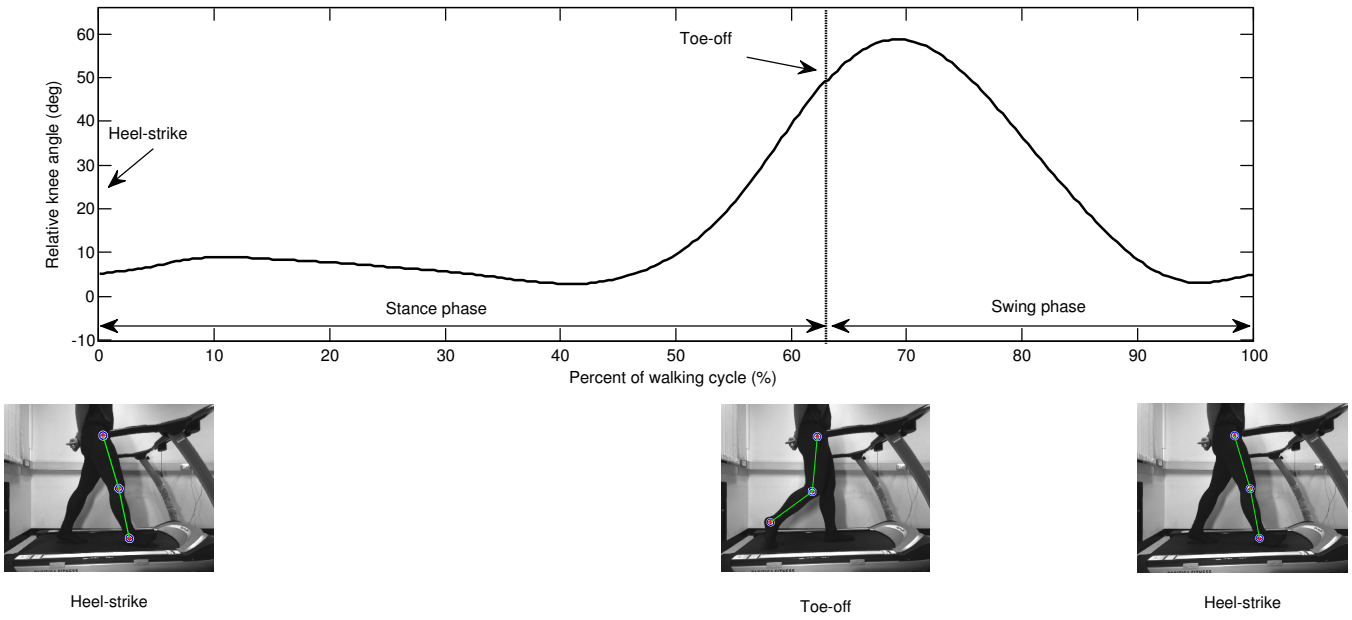


Fig. 8. The average knee angle at normalized time that is generated from the gait captured system at walking speed 3.6 km/h. Swing phase starts at toe-off until heel-strike.

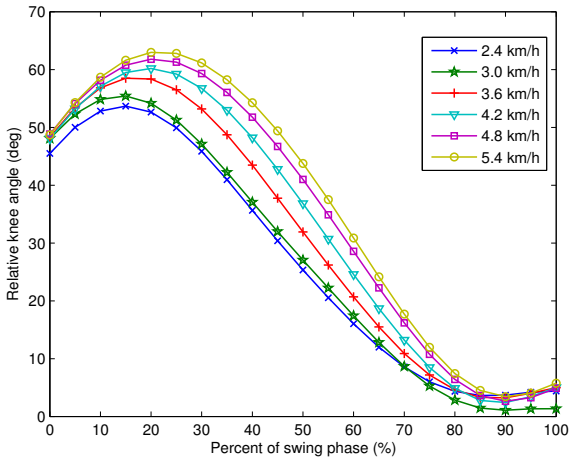


Fig. 9. The average knee angles in swing phase capture from the participant walking at various speeds.

consists of two neural network such as a hysteresis model and the gain function to estimate damping force from three inputs such as piston velocity, piston acceleration and applied current [19]. The modification has been changed in two points: (i) applied current has been replaced with applied voltage and (ii) the applied voltage input is filtered by the first order lag filter in order to apply to the model for simulating a response time of MR damper. The filtered applied voltage (v_{filter}) is calculated by the first order lag filter following equation:

$$v_{\text{filter}}(t) = c \cdot v(t) + (1-c) \cdot v_{\text{filter}}(t-1), \tag{9}$$

where v is the applied voltage and c is a constant value identified from data of the experimental system. The experimental system for collecting MR damper data is shown in Fig. 10. The modified EHM-based FNN was identified with data of an actual MR damper, Lord RD-8040-1, which has a stroke range of 54.8 mm and can generate a maximum force of 2447 N (peak-to-peak). The experimental setup used to capture MR damper data has been described in [20]. The MR damper was attached to a load cell that can measure both tension and compression forces. A servo motor was used as an actuator to generate excitations for the system. The MR damper piston motion varies between ± 14 mm and has a maximum speed of 180 mm/s. The National Instrument cRIO-9074 was used as a system controller. It contains a 400 MHz processor and 2M FPGA that can be programmed with LabVIEW to develop a real-time controller that generated a command voltage for the MR damper and recorded data from the system at a sampling time of 10 ms. The voltage range for the MR damper was 0 to 5 V, producing a current of approximately 0 to 1 A. Training process of EHM-based FNN is the same as the process describes in [19]. The c value was identified from optimization process with result of approximately 0.105.

D. Neural network predictive control implementation

1) Training the swing phase model: The swing phase model is the first part of NNPC, which is constructed using FNN as described in Section III-A. In the simulation, the control interval is selected as 20 ms. The training dataset includes two inputs, which are knee angle and command voltage, and the future knee angle is considered as the output. In order to generate the training dataset, the hip trajectory including θ_T, a_{hx} and a_{hy} are the actual data from the normal gait cycle while the applied voltage is randomized between 0 to 5 V and changed every 20 ms. The training dataset contains 200 swing data for slow, medium and fast walking speeds (2.4, 3.6 and 5.4 km/h). Thus, a total of 600 swing inputs and its targets, the future knee angle, are generated by the prosthetic knee simulation. The dataset is divided randomly into three groups,

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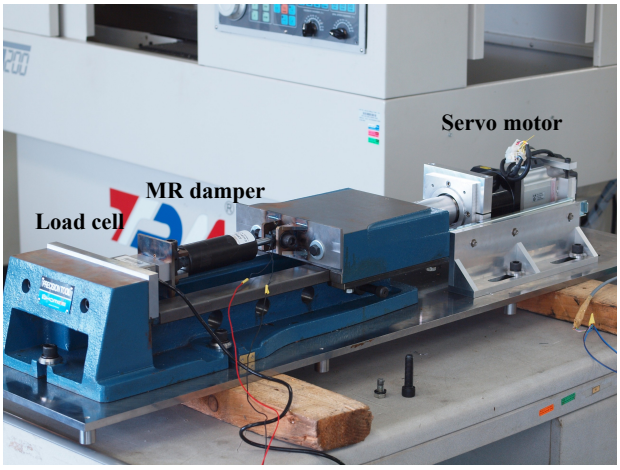


Fig. 10. The MR damper experimental system.

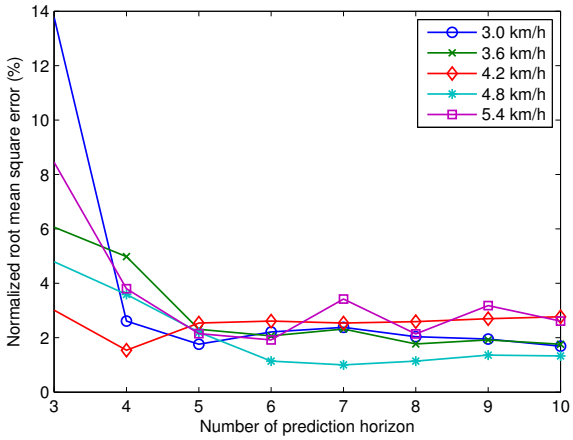


Fig. 11. Performance comparisons of NNPC with different numbers of prediction horizon.

consisting of training data, validation data and testing data with a 60:20:20 ratio. The Levenberg-Marquardt algorithm [21] is applied for training, while the performance is measured by the mean square error method. The training process will stop when the results of the validation checks do not improve significantly within six iterations or it reaches the maximum number of iterations, which is set to be 3000. The number of hidden nodes (n_{sw}) is determined by a training process that varies the number of nodes from 1 to 15. The node number estimation process runs 20 times with different initial weights and random data division. The lowest number that can adequately represent the training dataset will be selected as the best number of hidden nodes in order to avoid generalization errors. From the simulation, a suitable node count for the swing phase model is determined as 12. After the swing phase model is trained, it is implemented with C language using the same weights and biases as the identified FNN in order to perform at faster speed. The simulation was executed on a personal computer running Windows 7 with Intel i7-3.4 GHz processor and 8 GB of RAM. The calculation time for each evaluation can be measured approximately as 5.145×10^{-6} seconds when the source code is implemented using C language. On the other hand, the source code implemented with MATLAB m-file requires a calculation time of approximately 1.32×10^{-2} seconds. Thus, the swing phase model implemented with C language runs 2500 times faster than when implemented using MATLAB m-file.

2) Identifying parameters of the predictive control: The prediction horizon is determined by a preliminary test that simulates NNPC with various numbers of prediction horizon and using a control interval of 20 ms. The NNPC in this preliminary test employs the same structure as the proposed controller, but optimization is performed using the *fmincon* function based on trust-region method for nonlinear minimization [22], which is embedded in the Optimization Toolbox of MATLAB. The difference between desired knee angle and generated knee angle is calculated to indicate the performance of the control method using normalized root mean square error (NRMSE) method. The NRMSE can be calculated by the following equation:

$$NRMSE = \frac{\sqrt{\frac{1}{n_s}\sum_{t=1}^{n_s}[\theta_{K(\text{desired})}(t) - \theta_K(t)]^2}}{\max(\theta_{K(\text{desired})}) - \min(\theta_{K(\text{desired})})} \quad (10)$$

where θ_K and $\theta_{K(\text{desired})}$ are the generated and desired knee angle for each 1 ms simulation sample time. $\max(\theta_{K(\text{desired})}$ and $\min(\theta_{K(\text{desired})})$ are the maximum and minimum values of desired knee angle. n_s is the number of samples in the dataset. The preliminary results are shown in Fig. 11. The results show that a prediction horizon (n_{ph}) of 5 provides the best results at most walking speeds. However, this research chose $n_{ph} = 4$ to reduce the search space and achieve faster computation time. The weighting coefficients in the cost function (5) are selected as $w_1 = 0.1$, $w_2 = 0.2$, $w_3 = 0.3$, $w_4 = 0.4$ to give greater priority to latter horizon events.

3) Identifying parameters of particle swarm optimization: The parameters used for PSO are shown in Table II. The number of dimensions must be the same as the size of the prediction horizon. The weighting coefficients were chosen based on previous researches, where it was established that the solution converges faster than other values for five nonlinear functions [23]. Several swarm sizes (n_{swarm}) were tested for a walking speed of 3.6 km/h, as shown in Fig.12. The results indicate that PSO achieves fastest improvements for a swarm size of 5, with the best result found by a swarm size of 4. In NNPC, calculation time is critical because the control interval is only 20 ms. For this reason, a swarm size of 5 is selected. The results also show that 300 evaluations is enough for PSO, then the maximum iteration number is selected as 60. Since PSO is a meta-heuristic optimization method, it will generate different solutions during each run. Therefore, the proposed controller is validated 10 times and the average value is selected as the result.

E. User-adaptive control

The performance of NNPC is compared to user-adaptive control [8], which is a control strategy for variable-damping

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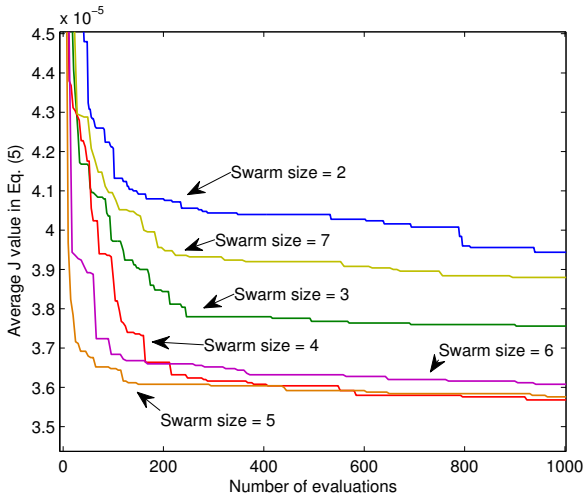


Fig. 12. Performance improvements of PSO with different number of swarm size. The number of evaluations are calculated by multiplying swarm size with number of iterations.

TABLE II
PARAMETERS FOR PARTICLE SWARM OPTIMIZATION.

Parameter	Value
number of dimensions (n_{dm})	4
population	5
maximum iterations	60
w_v	0.729
c_1	1.49445
c_2	1.49445

prosthetic knees with an adaptive mechanism for different walking speeds, and has been shown to achieve biologically realistic gaits during clinical trials. It is an open-loop control algorithm that provides different damping forces during swing flexion and extension. Knee damping can be increased or decreased using information from previous steps, allowing the controller to eventually converge to an optimal set of values. For comparison purposes, the user-adaptive controller that produces the best possible knee trajectory at each walking speed is determined by optimization. The genetic algorithm provided in the Optimization Toolbox of MATLAB is applied 10 times, and the best result is chosen as the user-adaptive controller that will serve as a benchmark for the proposed NNPC.

V. RESULT AND COMPARISON

The results of NNPC and user-adaptive control are shown in Table III and Fig. 13. The NNPC result is the mean value of 10 runs with standard deviation in the range of 0.031% and 0.22% for all walking speeds. They demonstrate that the swing phase controller using NNPC outperforms the user-adaptive control for any ambulation speed with good performance and error less than 1% except at 5.4 km/h. The results of user-adaptive control show that the knee trajectories reach the mechanical limit of the knee joint and terminate at 0 degree for all speeds, except at 2.4 km/h. This is different from the normal gait, leading to an asymmetrical walking pattern. In contrast, NNPC generates knee trajectories that are closer to the normal gait

TABLE III
RESULTS OF THE SWING PHASE SIMULATION.

Walking Speed (km/h)	NRMSE(%)	
	User-adaptive control	NNPC
2.4	2.70	0.81
3.0	2.24	0.88
3.6	3.65	0.61
4.2	3.29	0.80
4.8	3.18	0.71
5.4	3.46	2.42

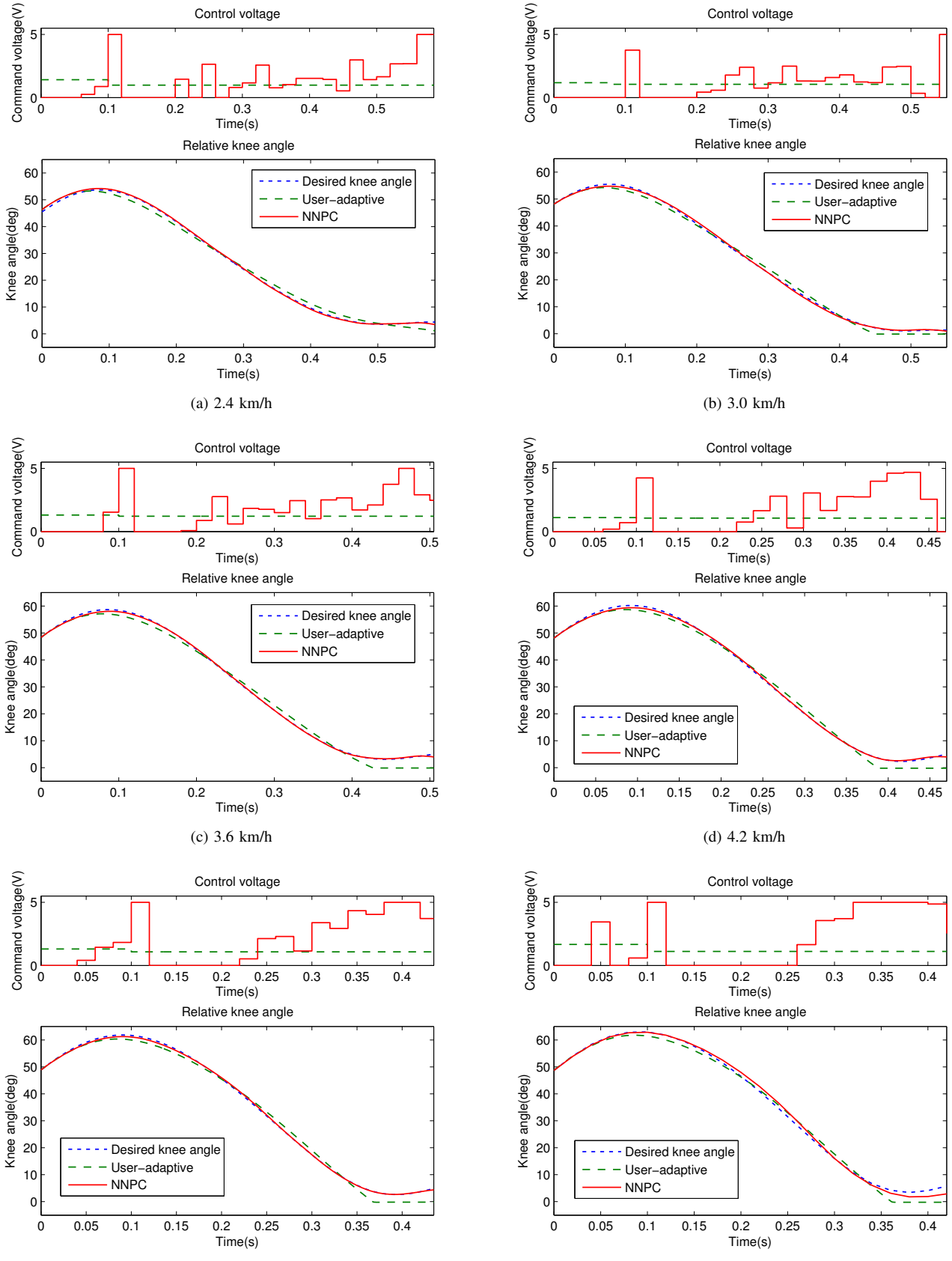
during all periods of the swing cycle, thus providing better gait symmetry. This is because NNPC has the ability to determine a suitable voltage during each control interval, whereas user-adaptive control is limited to only two constant damping levels that do not permit precise force generation. Though increasing the swing phase division can improve the performance of user-adaptive control, it remains difficult to find a suitable voltage for various ambulation speeds. Moreover, the user-adaptive control is an open-loop control algorithm that uses the information acquired from previous cycles to adjust the knee damping value. Thus, it cannot adjust the voltage command signals immediately in the case of an unexpected event or perturbation. On the other hand, NNPC is a close-loop control algorithm that provides adaptation ability as the environment changes.

The main robustness issues of NNPC relate to measurement noise and parameter uncertainties. The proposed controller measures the knee angle directly from an encoder installed at the knee joint, which provides a digital, noise-free signal and does not require additional filters. The effects of noise on the performance of NNPC is therefore negligible. Of greater concern are problems caused by inaccurate, uncertain, or variable system parameters, such as physical attributes and operating conditions of the prosthetic knee. A common scenario is a change in footwear or trousers that can lead to variations in weight and moment of inertia of the leg. For example, mens shoes can range from 0.1 to 0.5 kg per side. To simulate this, NNPC is trained assuming the weight of the leg is 3.0 kg, while validation is performed for an actual weight of 3.0 $\pm$ 0.5 kg at a walking speed of 4.8 km/h. The results show that NNPC generate different command signals for each situation that produce knee trajectories with NRMSE of 1.19%, 0.62%, and 0.69%, when the leg is 2.5, 3.0, and 3.5 kg, respectively, as shown in Fig. 14. The difference in performance is negligible and it can thus be concluded that NNPC is robust to parameter estimation errors and changes in system variables.

Though these results all indicate the high performance and robustness of NNPC when compared to existing algorithms, like user-adaptive control, it is noted that calculation of the evaluation function during optimization requires more computation time. This raises a new concern about whether NNPC can be implemented on an actual embedded system. To answer this question, the performance of NNPC has been evaluated on a microcontroller. Due to the size of a typical prosthetic leg, only controller boards with a small-size and good power efficiency are considered as appropriate. In this work, ARM

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(a) 2.4 km/h

(b) 3.0 km/h

(c) 3.6 km/h

(d) 4.2 km/h

(e) 4.8 km/h

(f) 5.4 km/h

Fig. 13. Voltage command and generated knee angle obtained from simulations of 6 ambulation speeds. Blue dotted line ('··') denotes the desired knee angle, green dashed line ('- -') denotes results of the user-adaptive control and red solid line ('-') denote results of NNPC.

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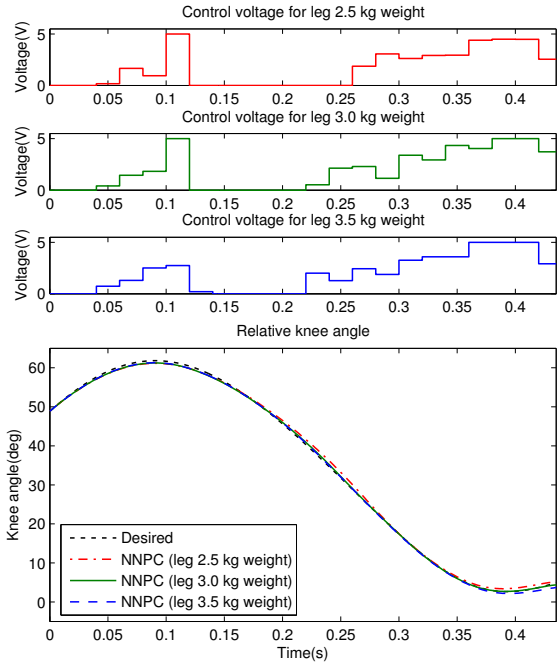


Fig. 14. Voltage command and generated knee angle obtained from NNPC at 4.8 km/h when the weight of the leg is varied between 2.5, 3.0, and 3.5 kg.

TABLE IV
AVERAGE COMPUTATION TIME OF NNPC RUNNING ON ARM CORTEX-A8 WITH 1 GHZ

	Times (ms)
Time for each evaluation	0.0344
Time for 300 evaluations	10.308
Time for PSO process	0.773
Total Time	11.081

Cortex-A8 installed in a BeagleBone Black board is used to evaluate the speed of NNPC. It operates at 1 GHz, with 512 MB RAM and runs the Linux operating system. Moreover, it includes special hardware accelerator for floating-point calculation. During the testing process, NNPC is programmed using C language and compiled with the GNU compiler. Experimental data is collected from 3.6-km/h dataset and the parameters of PSO is similar to those shown in Table II. NNPC runs for 1,000 rounds and the average time usage during the test is measured. As shown in Table IV, the average time for calculating NNPC is 11.081 ms, which is approximately 55% of the control interval. Also, the evaluation function and PSO consume 93% and 7% of NNPC calculation time, respectively. This results show that FNN takes up most of calculation time; thus, a microcontroller suitable for implementing NNPC algorithm should have the ability to quickly calculate floating-point to achieve good control quality. This experiment suggests that it is possible to run NNPC on microcontrollers with an appropriately fast processor.

VI. CONCLUSION

NNPC is proposed to control a semi-active prosthetic knee during the swing phase and is tested at various walking speeds. It consists of a swing phase model constructed with FNN and a predictive controller optimized by PSO. NNPC solves the problem of MR damper that provides only damping force, while normal gait requires both active and damping force. Unlike the active knee implementation, The MR damper cannot generate desired knee torque directly because of its passive behavior and nonlinear dynamic characteristics. NNPC provides a predictive ability of the next knee trajectory for a given voltage, then optimizes the input voltage until a good result is found. The best voltage will be used as the input command to the MR damper. The simulation shows that NNPC can mimic knee trajectory of a normal gait and generates better results compared to the user-adaptive control. It is also demonstrated that NNPC is robust to variation in system parameters and can be implemented on real-time embedded platforms.

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